Procedure Drift Finder — Team Playbook

AMD Developer Hackathon: ACT III · prepared 10 Oct 2026 · Track 1 Intelligent Industry. Facts come from public pages (links at the end). Scores and opinions are judgment, so verify before the pitch.

1. The project in 60 seconds (paste this into any new chat to restart context cheaply)

- **Idea:** a tool that reads the notes a plant already writes (handover notes, near-miss reports, maintenance logs) against its written SOPs, finds drift (practice that differs from the procedure) and shows the exact quote as proof.
- **User:** the plant safety or process manager. **Track:** 1, Intelligent Industry.
- **Pipeline:** load SOP steps → load notes → match each note to likely steps → AI judge answers in a fixed format (step, drift type, exact quote, confidence) → plain code checks the quote really appears word for word, otherwise the flag is dropped → count flags by step, shift and week → a human approves or rejects → draft an SOP fix → benchmark against planted drifts.
- **Drift types:** skipped step · reordered steps · workaround or substitution · SOP outdated or impractical · SOP unclear.
- **Golden rules:** flag only when a note affirmatively says something (silence is not evidence) · report on steps and shifts, never people · say upfront the data is synthetic · never claim first.
- **Team (6):** 2 front end, 2 back end, 1 marketing, 1 flex. **Stack:** Streamlit, vLLM on AMD Developer Cloud (MI300X), Evolus (optional partner), Docker.
- **Claude Code note:** it needs a paid plan or Console credits. Free Claude chat is the fallback.

2. Prizes we are aiming for

Prize	What it needs	Status
Main: Track 1 Intelligent Industry	A real manufacturing problem, a working product (not only a dashboard), results operators understand, a measurable improvement. AMD must run a meaningful part of the working product shown to judges.	Core plan
Evolus Business Agents	Build in an Evolus event workspace; main agent uses an open model on our own vLLM endpoint on AMD Developer Cloud; at least 2 Evolus building blocks; app drives Evolus through its APIs; containerized app, public repo, README that runs. Evolus checks workspace configuration and usage statistics.	Planned, with a gate on 13 Oct
Vibe Generation (PwC) special prize	Judged on Hybrid (strategy, quantified value, commercial attitude), Becreatives (technical execution, orchestration of AI tools, ability to ship) and Vibe Generation (taste, originality, human plus AI workflow).	Fits our human-review loop. Needs polish and numbers
Google	The page says Google technology is optional and a project can win the main prize plus partner prizes. **No rules are shown.**	Unknown. ACT II's Google prize rewarded Gemma, including Best AMD-Hosted Gemma Project. We test Gemma anyway

Priority order: 1) core + AMD proof + benchmark, 2) Evolus, 3) Google. Eligible is not the same as winning, and a partner prize must never delay the core.

Unknowns to resolve: whether we may enter more than one track ('choose one or more tracks' versus a single 'Selected challenge' field), the Google rules, and the exact deadline (older copies of the page said online build 12–17 Oct; the current one says 12–18).

3. What earlier AMD winners teach us

Facts

- **AMD Developer Hackathon (May 2026):** 481 projects, 2,618 teams. Champion: Chaos Economy, a multi-agent reinforcement-learning market simulator. Runner-up: Strike Lab. Third: Automato.
- **ACT II (July 2026):** 1,149 projects, 4,885 teams averaging 1.6 members. Champion: Intelligence for Existing CCTV Systems (team ROC-Dragon, Unicorn track). Runner-up: OyeHoye Zero Tokens. Third: CacheForge, a topology-aware KV cache.
- **The ACT II champion** turned cameras companies already own into a system you can question in plain language, returned verified answers with video evidence, tracked people across 28 cameras, and ran a multi-model pipeline (RT-DETR, BoT-SORT, OSNet, Gemma) on one MI300X using ROCm, MIGraphX and vLLM-ROCm. It claims hours or days of review shrink to minutes.
- **248 ACT II submissions did not qualify** for automated scoring, mostly because the container image could not be pulled (not public or wrong reference), the container crashed (it must run on linux/amd64), timeouts, or accuracy-gate failures.
- Gemma was one of the most used technologies there (325 submissions).

Caveat: I only saw project descriptions and leaderboards, not the judges' notes. These are patterns, not proof of what scored.

What winners did	What we do
Upgraded something the customer already owns	We upgrade the notes the plant already writes: no cameras, no new monitoring
Verified answers with evidence	Code-checked quotes and an evidence viewer are our hero feature
Deep, visible AMD use	Two models on one MI300X; show notes per minute and cost per 1,000 notes; name ROCm and vLLM on the architecture slide

What winners did	What we do
Hard numbers and a time-saved story	Benchmark on planted drifts plus measured manual-versus-tool review time
Efficiency counted (Zero Tokens and a KV-cache project placed top 3)	We report tokens and throughput honestly
Submission mechanics sank many teams	Container tested on linux/amd64 by 15 Oct, public repo, README that runs

4. The feature set, step by step

Must = no entry without it. **Should** = what makes us a contender. **Stretch** = only if ahead.

ID	Feature	Why judges care	Owner	Done when
F1	SOP loader: split each SOP into numbered steps	Everything keys off step numbers	Data lead + Builder A	3 SOPs, 8–15 steps each, load in one call
F2	Notes loader: about 200 notes (date, shift, source, text)	Realistic input and volume	Data lead	notes file plus answer key
F3	Matcher: link each note to candidate steps with a small fast model or embeddings	Cheap, fast, shows two models on one GPU	Builder A	top 3 candidate steps per note
F4	Judge: larger model returns a fixed format (step, drift type, exact quote, confidence) and flags only what the note affirmatively states	The core intelligence and the useful recommendation Track 1 asks for	Builder A	valid format on every call (retry when malformed)

ID	Feature	Why judges care	Owner	Done when
F5	Quote verifier in plain code: the quote must appear word for word in the note	Our trust feature. Never cut	Builder A	every shown flag verified; dropped count reported
F6	Review queue: approve or reject each flag	Human plus AI workflow; safety rule	Front end	decisions saved
F7	Heat strip: flags by SOP step, week and shift, with trend direction	Results operators understand	Front end + Builder B	click a cell, see its flags
F8	Benchmark: precision, recall, quote-verified rate, false alarms, notes per minute against 25 planted drifts, plus a Where we are wrong page	Measurable improvement and honesty	Builder B + Data lead	numbers on screen from the real run
F9	AMD proof panel: model names, GPU, endpoint, throughput, cost per 1,000 notes	AMD must appear in the demo	Builder A + Front end	live call shown in the video
F10	Container and submission hygiene: Dockerfile for linux/amd64, public repo, README that runs, keys in environment variables	Evolus rule and ACT II's main failure mode	Lead + Front end	clean-machine test passes

ID	Feature	Why judges care	Owner	Done when
F11	SOP-fix drafts for 'outdated or impractical' and 'unclear' flags, human approved	Drift is not always a violation	Builder A	one drafted fix per such flag
F12	Evolus integration: document extractor (SOP and logbook PDFs → structured data) plus a workflow with sub-agents, called from our app by API	Evolus prize	Builder A + Builder B	2+ building blocks used with real API calls
F13	Evidence viewer: click a flag, see the SOP step, the highlighted quote and similar notes on other shifts	Our version of verified answers with evidence	Front end	one screen tells the whole story
F14	Value calculator: minutes to review 200 notes by hand versus with the tool, plus a sourced cost of one incident	Hybrid criterion and Business Value	Marketing	one slide with measured numbers
F15	Gemma test: run Gemma as one candidate judge on AMD	Cheap insurance for a Google prize	Builder A	precision comparison table
F16	Stretch: voice handover notes to text with Whisper on AMD	Realistic input	Builder B	one demo clip
F17	Stretch: notes in a second language	Uses the team's languages	Data lead	20 notes, results reported

ID	Feature	Why judges care	Owner	Done when
F18	Stretch: alert when a step's weekly drift count rises	Operator value	Builder B	one alert in the demo

5. Requirement checklists

Track 1 (from the page)

- Real manufacturing problem → drift and normalization of deviation (cite accident inquiries with sources in the pitch).
- Working product → F1–F9 running end to end on the AMD GPU.
- Results operators understand → heat strip and evidence viewer.
- Measurable improvement → time to review notes and detection numbers.

Evolus (all five must be true): workspace created with the kickoff event code (free, up to 6 members, 50,000 credits until 31 Oct) · main agent uses our open model through our own vLLM endpoint on AMD Developer Cloud · at least two Evolus building blocks · our app drives Evolus through its REST API · containerized app, public GitHub repo, README that runs, no keys in the repo.

Vibe Generation: Hybrid → F14 and a clear buyer · Becreatives → F3 to F5 and F12 working together and shipped · Vibe Generation → clean design and a story about humans and AI working together.

AMD rule: the AMD part must be in the demo shown to judges.

6. Roles

Role	Who	Owns
Lead	Decide today (doubles with another role)	Scope, deadlines, final calls
Data lead	Flex person	SOPs, notes, answer key, planted drifts (the critical path)
Builder A	Back end 1	AMD GPU, vLLM, F3 to F5, F11, F12, F15

Role	Who	Owns
Builder B	Back end 2	Aggregation, benchmark, F7 data, F12 workflow
Front end x2	Two builders	F6, F7, F9 panel, F13, container (see the Front-End Guide)
Pitch and marketing	Marketing	README, slides, video, F14, Discord questions, interviews

7. Schedule and gates

Date	Goal
Oct 10–11	Everyone signs up (lablab, Discord, AMD AI Developer Program) and claims credits. Agree the data format. Front end builds on fake data. Post the Discord questions.
Oct 12	Test the AMD GPU with vLLM and two models. Create the Evolus workspace with the kickoff code, connect the endpoint, run one workflow on one note. Create the repo.
Oct 13	SOPs, notes, answer key. First pipeline on 10 notes. **Gate (evening):** if Evolus cannot call our endpoint, drop it and continue direct.
Oct 14	Quote verifier, drift types, all 200 notes. Build the container once as a test.
Oct 15	Aggregation, review queue, evidence viewer, SOP-fix drafts. Swap fake data for real output. Clean-machine container test.
Oct 16	Benchmark, fix the worst errors, freeze features.
Oct 17	Polish, record the demo, submit early.
Oct 18	Buffer. Confirm the exact deadline.

Cut order if behind: stretch features → F15 → F11 → F12 (only if the gate failed) → trend charts. **Never cut:** F5 quote verifier, F8 benchmark and answer key, AMD proof, container.

Design rule: keep one function, judge_note, that can call either Evolus or our vLLM endpoint directly, so we can flip a switch if Evolus fails.

8. Test data plan (the most important job)

- Write 3 to 5 SOPs (for example lockout-tagout, forklift battery charging, confined-space entry).
- Plant 25 known drifts, each as a one-line fact.
- Generate about 200 notes in varied styles: some with drifts, some clean, some traps (same words, procedure followed).
- **Avoid circular testing:** a different model family writes the notes than the one that judges them, and teammates hand-write about 30 notes.
- Keep an answer key listing which note contains which drift.
- Say clearly in the demo that the data is synthetic. Add public incident narratives only where access is verified.

9. Rules we follow

- Flag only what a note says; never infer from silence.
- Report on steps and shifts, never individuals; no names in the interface.
- Every flag needs a human decision; we are not a safety control.
- No keys or endpoint addresses in GitHub.
- Honest numbers, including false alarms.

10. Questions for Discord (Lead or marketing posts today)

1. What does 'choose one or more tracks' mean for the Selected challenge field?
2. What are the Google partner prize requirements?
3. Does the Vibe Generation prize have extra technology requirements?
4. How should we expose a vLLM endpoint to Evolus securely?
5. Is clearly labelled synthetic data acceptable for Track 1?
6. What is the exact submission deadline and time zone?

11. Submission checklist and demo story

Title · short and long descriptions · track and technology tags · cover image · demo video · slides · public GitHub repo with a README that runs · app URL · container builds for linux/amd64 and starts · no keys committed.

3-minute demo: 1) the gap: one SOP step and one contradicting note (20s) · 2) the load: 200 notes on the AMD GPU, notes per minute (20s) · 3) the result: heat strip, click the top drift, see the step and quote (60s) · 4) the twist: an 'SOP is impractical' flag and its drafted fix (30s) · 5) the proof: benchmark against 25 planted drifts (30s) · 6) honesty slide: where our AI is wrong (20s).

Sources

- Hackathon page: https://lablab.ai/ai-hackathons/amd-developer-hackathon-act-iii
- ACT II results: https://lablab.ai/ai-hackathons/amd-developer-hackathon-act-ii/live
- First AMD hackathon results: https://lablab.ai/ai-hackathons/amd-developer/live
- ACT II champion project: https://lablab.ai/submissions/l85sr7g6ahkq4vhf9gk2na4j
- Claude Code quickstart: https://code.claude.com/docs/en/quickstart